PROGRAMME **:** Diploma Programme in Computer Technology
COURSE **:** Computer Security(COS) **COURSE CODE :** 232406

LEARNING AND ASSESSMENT SCHEME:

<table>
<tr><th colspan="5">Learning Scheme</th><th rowspan="3">Credits</th><th colspan="11">Assessment Scheme</th><th rowspan="3">Total Marks</th></tr>
<tr><th colspan="3" rowspan="2">Actual Contact Hrs./Week</th><th rowspan="3">SLH</th><th rowspan="3">NLH</th><th rowspan="3">Paper Duration</th><th colspan="4" rowspan="2">Theory</th><th colspan="4">Based on LL and TSL</th><th colspan="2" rowspan="2">Based on Self Learning</th></tr>
<tr><th colspan="4">Practical</th></tr>
<tr><th>CL</th><th>TL</th><th>LL</th><th></th><th>FA-TH</th><th>SA-TH</th><th colspan="2">Total</th><th colspan="2">FA-PR</th><th colspan="2">SA-PR</th><th colspan="2">SLA</th><th></th></tr>
<tr><td rowspan="2">03</td><td rowspan="2">00</td><td rowspan="2">02</td><td rowspan="2">01</td><td rowspan="2">06</td><td rowspan="2">03</td><td rowspan="2">03</td><td>Max</td><td>Max</td><td>Max</td><td>Min</td><td>Max</td><td>Min</td><td>Max</td><td>Min</td><td>Max</td><td>Min</td><td rowspan="2">150</td></tr>
<tr><td>30</td><td>70</td><td>100</td><td>40</td><td>25</td><td>10</td><td>--</td><td>--</td><td>25</td><td>10</td></tr>
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IKS Content: Total Learning Hours for Term: 00 Hrs
# Indicates Online Examination, @ Internal assessment

1.0 RATIONALE:

The Computer Security study equips students with essential skills to protect digital assets and data integrity. It addresses various threats and vulnerabilities, ensuring students understand how to implement effective security measures. The curriculum emphasizes practical applications, including network security and cryptography, to prepare students for real-world challenges. By focusing on both theoretical concepts and hands-on experience, it aims to build a strong foundation in safeguarding information systems. This knowledge is critical as cyber threats continue to evolve and impact various industries.

2.0 INDUSTRY / EMPLOYER EXPECTED OUTCOME:

The aim of the course is to help the student to attain the following industry /employer expected outcome: students should effectively identify, analyze, and mitigate security threats, ensuring robust protection of digital assets and systems.

3.0 COURSE OUTCOMES:

The course content should be taught and learning imparted in such a manner that students are able to acquire required learning outcome in cognitive, psychomotor and affective domain to demonstrate following course outcomes:

1. Describe security threats, attacks, risks, challenges, and security basics of computer systems.
2. Prepare and implement cryptographic algorithms and certificates to secure sensitive information.
3. Explain standard practices and protocols to secure network and networking resources.
4. Implement OS and application hardening techniques to secure OS and applications.
5. Apply various types of recovery techniques and cyber crimes to avoid malpractices.

4.0 COURSE DETAILS:

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
Unit-I **Computer and Operational Security**	1a. Understand security and security threats 1b. Explain different types of attacks and basic principles of security 1c. Describe different roles of people to maintain security in organization 1d. Knowledge about different physical security mechanisms	1.1 Security: Introduction, need for security, Threats to security, Avenues of attack, and Steps in attack. 1.2 Types of attack: Denial of service, backdoors and trapdoors, sniffing, spoofing, man in the middle, replay, TCP/IP Hijacking, Encryption attacks, Malwares, Viruses, Logic bombs, and Trojan	11

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
		horses. 1.3 Security Basics: Confidentiality, Integrity, Availability, Operational model of Computer Security, Layers of security. 1.4 Role of people in security: Password (selection, Management, Components of good password), Piggybacking, Shoulder surfing, Dumpster diving, Installing unauthorized software /hardware, Access by non employees, Security awareness, Individual user responsibilities, Security policies, standards, procedures and guidelines. 1.5 Physical Security: Access Control (DAC, MAC, RBAC), Authentication, Biometrics (finger prints, hand prints, Retina, patterns, voice patterns, signature and writing patterns, keystrokes), Social Engineering.	
Unit-II **Information Security**	2a. Indentify and explain different types of security algorithms 2b. Describe symmetric and asymmetric cryptographic techniques	2.1 Introduction: Cryptography, Cryptanalysis and Cryptology, Substitution techniques: Caesar's cipher, mono-alphabetic and Poly-alphabetic, one-time pad, Transposition techniques: Rail fence technique, simple columnar, Steganography, Hashing. 2.2 Symmetric and asymmetric cryptography: Introduction to Symmetric encryption, DES (Data encryption Standard) algorithm, Asymmetric key cryptography: Digital Signature. 2.3 Public key infrastructures: basics, digital certificates, certificate authorities, registration authorities, Trust models (Hierarchical, peer to peer, hybrid)	09

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
Unit-III **Network Security & Intrusion Detection System**	3a. Introduction to different security mechanism for network security 3a. Understand intrusion and intrusion detection system	3.1 Firewalls: Need for Firewall, limitations, characteristics, Types of Firewall : Hardware, Software, Packet filter, Proxy Server, Hybrid, Application gateways, circuit level gateway, Implementing Firewall. 3.2 Virtual Private Network, Kerberos, security topologies: security zones, DMZ, Internet, Intranet, VLAN. 3.3 Email security: Email security standards, working principle of SMTP, PEM, PGP, S/MIME. 3.4 IP security: overview, architecture, IPSec Configuration, IPSec Security. 3.5 Intrusion Detection: Intrusion detection systems (IDS), host based IDS, network based IDS, Honey pots.	09
Unit-IV **Software & Web Security**	4a. Identify different techniques to secure OS and applications 4b. Apply steps to secure software, applications, and OS	4.1 Operating system security: Operating system hardening, general steps for securing operating system, updates: hotfix, patch, service pack. 4.2 Application Security: Application hardening, application patches, secure code techniques, buffer overflows, code injection, least privilege, good practices. 4.3 Web security threats, web traffic security approaches, Secure Socket layer and transport layer security.	08
Unit-V **Recovery Techniques & Cyber Crime**	5a. Understand basics of recovery and cyber crimes 5b. Implement different recovery techniques to recover sensitive information 5c. Describe different types of cyber crimes and law to prevent such crimes	5.1 Recovery: Introduction to Deleted File Recovery, Formatted Partition Recovery, Data Recovery Tools, Data Recovery Procedures and Ethics. 5.2 Cyber Crimes: Introduction, Hacking, Types of Hacking, Cracking, Pornography, Software Piracy, Intellectual property, Legal System of Information Technology, Mail Bombs, Bug Exploits, Cyber Crime Investigation, Ethical Hacking. 5.3 Cyber Laws: Introduction to IT	08

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
		act 2000 and IT act 2008s.	
Total			**45**

5.0 SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):

Unit No.	Unit Title	Distribution of Theory Marks			
		R Level	U Level	A and above Levels	Total Marks
I	Computer and Operational Security	08	06	04	18
II	Information Security	06	06	02	14
III	Network Security & Intrusion Detection System	06	04	04	14
IV	Software & Web Security	04	04	04	12
V	Recovery Techniques & Cyber Crime	04	04	04	12
	TOTAL	**28**	**24**	**18**	**70**

6.0 LABORATORY LEARNING OUTCOME AND ALLIED PRACTICAL/ TUTORIAL EXPERIENCES:

The tutorial/practical/assignment/task should be properly designed and implemented with an attempt to develop different types of cognitive and practical skills **(Outcomes in cognitive, psychomotor and affective domain)** so that students are able to acquire the desired programme outcome/course outcome.

Note: Here only outcomes in psychomotor domain are listed as practical/exercises. However, if these practical/exercises are completed appropriately, they would also lead to development of **Programme Outcomes/Course Outcomes in Affective domain** as given in the mapping matrix for this course. Faculty should ensure that students also acquire Programme Outcomes/Course Outcomes related to affective domain.

Sr. No.	Unit No.	Laboratory Learning Outcome	Laboratory Experiment Title	Hours
01	I	1a. Apply basic concepts of security.	• Case study on : Role of people in security	02
02	II	2a. Install and configure open source software for cryptography. 2b. Apply simple Transposition Permutation Encrypt and decrypt techniques.	• Install open source Latest version of any open source software like Cryptool software and Encrypt and decrypt the message using Simple Transposition Permutation(Cryptool)	04
03	II	3a. Apply Encryption and decryption techniques.	• Encrypt and decrypt the message using Caesar Cipher With Variable Key (Cryptool)	04
04	II	4a. Implement DES algorithm.	• Write a simple program for DES encryption/decryption in java or C or .net	04
05	II	5a. Create Digital Signature document.	• Create Digital Signature document using open source tool like Cryptool	04
06	III	6a. Install and configure firewall.	• Install and configure of firewall and its policies	04

Sr. No.	Unit No.	Laboratory Learning Outcome	Laboratory Experiment Title	Hours
07	III	7a. Implement E-mail security.	• Trace email origin using any open source tool like eMail Trace Pro utility, Pretty Good Privacy with GnuPG.	02
08	IV	8a. Use the security provided with windows operating system(User authentication)	• Use the security provided with windows operating system(User authentication)	02
09	IV	9a. Set up and recover password of computer system☐	• Recover the password of windows OS using password recovery utility (John the ripper) or any other utility	04
10	V	10a. Trace the path of an website/ web server using open source tool like tracert utility	• Trace the path of an website/ web server using open source tool like tracert utility	02
Total				30

7.0 SELF LEARNING: (Assignment / Activities for Specific Learning / Skill Development / Online Courses / Micro projects.

Other than the classroom and laboratory learning, following are the suggested Student-related co-curricular activities. This can be undertaken to accelerate the attainment of the various outcomes in the course.

Each student must perform at **least one activity**. Following are suggested as sample assignments. Faculty is expected to assign individual assignment to each student and keep record in hard/soft copy as required for assessment.

Micro project

- User A wants to send message to user B securely on network.
    i. Select any two techniques to encrypt message.
    ii. Implement both the techniques.
    iii. Evaluate result of implementation.
    iv. Compare complexity of both techniques.
    v. Prepare report.
- Prepare admin level report of company who wants to implement allocate fixed system to each employee for authentic access to maintain security.
    i. Explain various single level authentication method available to access the system.
    ii. Analyse the weakness and security threats to this problem.
    iii. Suggest multi factor authentication for given problem situation.
    iv. Compare impact of single and multi-factor authentication on given situation.
- A bank has more than 1000 user accounts. Around 100 users received message regarding deduction of specific amount without intimation and after that all authorized user are not able to access online banking service of that bank.
    i. Identify type of crime and attack.
    ii. Write procedure to investigate that crime.
    iii. Write preventive measure to avoid such type of attack in future.
    iv. Write punishment of such type of attacks and state cyber law act.
    v. Write a report.

- Case study on Cyber Crime in Social Engineering in India.
    - i. Explain various Social Engineering attacks.
    - ii. Select topic for case study.
    - iii. Write problem statement of attack.
    - iv. Write procedure to investigate that attack.
    - v. Write a report

8.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):

These are sample strategies, which the teacher can use to accelerate the attainment of the various outcomes in this course,

1. Guide students in undertaking mini projects.
2. Demonstrate students thoroughly before they start doing the practice.
3. Encourage students to refer different website and YouTube channels to have deeper understanding of the subject.
4. Observe and monitor the performance of students in lab.
5. Use of online platforms like MOOCs, Swayam, Spoken tutorial, NPTEL, Infosys springboard etc.

9.0 ASSESSMENT METHODOLOGY

1. Formative assessment: TH
   Periodic Test: 30marks
2. Summative assessment: TH
   70 marks Final theory paper.
3. Formative assessment: PR
   Lab performance, Viva
4. Self learning includes micro projects /assignments/ other activities.

10.0 LEARNING RESOURCES:

A) Books:

Sr.No.	Author	Title of Book	Publication
1	Atul Kahate	Cryptography and Network Security	Tata McGraw Hill
2	William Stallings, Lawrie Brown	Computer Security Principles and Practices	Pearson Education
3	Dieter Gollman	Computer Security (Second Edition)	Wiley India Education
4	C K Shyamala, N Harini	Cryptography and Security	Wiley India
5	Matt Bishop	Introduction to Computer Security	Addison-Wesley

B) Software/Learning Websites:

1. https://www.geeksforgeeks.org/computer-security-overview/
2. https://intellipaat.com/blog/what-is-computer-security/
3. https://www.tutorialspoint.com/fundamentals_of_science_and_technology/cyber_crime_and_cyber_security.htm
4. https://www.techtarget.com/searchsecurity/definition/cryptography

C) Major Equipment/ Instrument with Broad Specifications:

Sr.No.	Equipment	Specification
1	Desktop Computer	PC Specifications to be followed: Processor: i3 or i5 (or Higher) RAM: 4 GB (or Higher) HDD: 1 TB SATA Monitor: TFT LCD OS: Genuine Windows 8 or 10 Professional or Home Premium or Windows 8 or 10 Ultimate Internet Connection
2	LCD Projector	Display Type: LCD Light Output: 3200 Lumens
3	Software	Open Source tools for Security, E-mail Security Tool. (Open-source tool), Any freeware password recovery tool, Any compiler (TurboC / Online 'C' compiler), Encryption and decryption tool. (Open-source tool), Web Browser. (Any Web Browser)

11.0 MAPPING MATRIX OF PO's, CO's and PSO's:

Course Outcomes	Programme Outcomes (PO's)							Programme Specific Outcomes (PSO's)		
	1	**2**	**3**	**4**	**5**	**6**	**7**	**1**	**2**	**3**
CO1	H	M	--	M	H	L	L	H	L	--
CO2	H	L	M	L	--	--	M	--	H	L
CO3	H	--	H	--	L	M	M	--	--	--
CO4	M	M	--	L	H	--	M	L	M	--
CO5	H	--	L	--	--	L	M	L	--	--

H: High Relationship, M: Moderate Relationship, L: Low Relationship.

12.0 SUGGESTED QUESTION PAPER PROFILE: